Advanced Topics in Computer Graphics I

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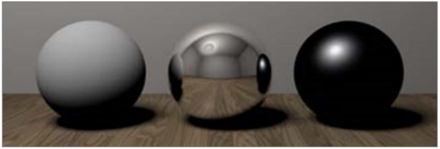
Path Guiding

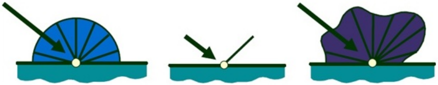
Path Guiding

Want to solve rendering equation

$$\boldsymbol{L}_o(\boldsymbol{x}, \boldsymbol{\Omega}_o) = \boldsymbol{L}_e(\boldsymbol{x}, \boldsymbol{\Omega}_o) + \int_{\mathcal{H}_i} \boldsymbol{f}_{BRDF}(\boldsymbol{\Omega}_i, \boldsymbol{x}, \boldsymbol{\Omega}_o)\, \boldsymbol{L}_o(\boldsymbol{h}(\boldsymbol{x}, \boldsymbol{\Omega}_i), -\boldsymbol{\Omega}_i) \cos(\theta_i)\, d\omega_i$$

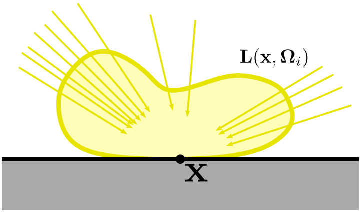
Problem: Typically, the BRDF's $\boldsymbol{f}_{BRDF}(\boldsymbol{\Omega}_i, \boldsymbol{x}, \boldsymbol{\Omega}_o)$ of the modeled materials are known. However, the illumination $\boldsymbol{L}_o(\boldsymbol{h}(\boldsymbol{x}, \boldsymbol{\Omega}_i), -\boldsymbol{\Omega}_i)$ at the surfaces is not known.



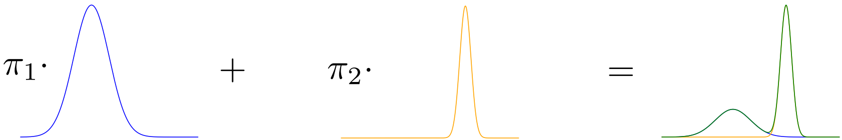


Idea: Use already computed samples of $\boldsymbol{L}_o(\boldsymbol{h}(\boldsymbol{x}, \boldsymbol{\Omega}_i), -\boldsymbol{\Omega}_i)$ to guide future samples (Vorba, Karlik, et al., 2014).

Question: How to model the incoming light distribution $\boldsymbol{L}_i(\boldsymbol{x}, \boldsymbol{\Omega}_i)$?



Idea: Use Gaussian mixture models to approximate distribution.



Multivariate Normal distribution (Gaussian)

A **Multivariate Normal distribution (Gaussian)** of dimension D is defined as

$$\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D \det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)$$

where the governing parameters are the mean $\boldsymbol{\mu}$ and the covariance matrix $\boldsymbol{\Sigma}$.

Gaussian Mixture Models

A mixture of Gaussians with K components is defined as

$$p(\boldsymbol{x}) = \sum_{k=1}^{K} \underbrace{\pi_k \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)}_{C_k}$$

where the mixing coefficients $\pi_k \in [0,1]$ are normalized:

$$\sum_{k=1}^{K} \pi_k = 1$$

Question: How to find the parameters of the GMM from a number of identically, independently sampled data points $\boldsymbol{X}$?
Idea: Find parameters $\boldsymbol{\theta}$ that maximize the probability for observing the dataset, i.e.,

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} p(\boldsymbol{X} \mid \boldsymbol{\theta})$$

In practice, this formulation is often rewritten as the minimization of the negative log-likelihood (NLL). In our case the NLL for a GMM is given as

$$-\log p(\boldsymbol{X}|\boldsymbol{\theta}) = -\sum_{i=1}^{n} \log\left\{\sum_{j=1}^{K} \pi_j \mathcal{N}(\boldsymbol{x}_i|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)\right\}$$

Minimize NLL by **Expectation-Maximization** (EM) algorithm

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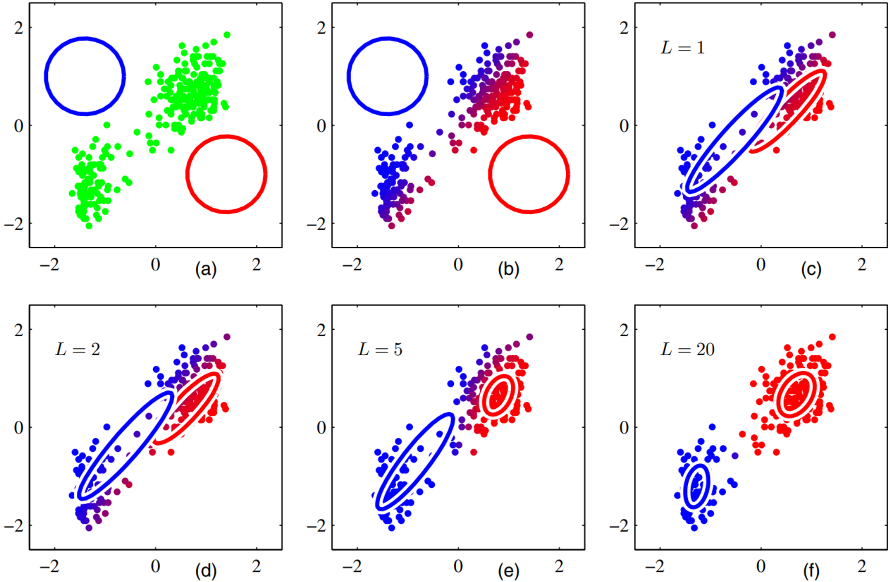
Minimize NLL by **Expectation-Maximization** (EM) algorithm.
General Idea:

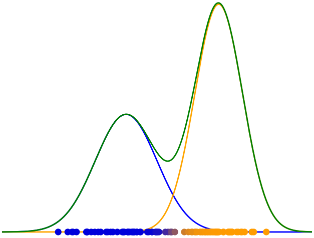
- **E(stimation)-Step**: Compute **responsibilities** r_{ij} that describe if a data point $\boldsymbol{x}_i$ corresponds to a gaussian component j.
- **M(aximization)-Step**: Re-estimate the parameters using the current responsibilities.

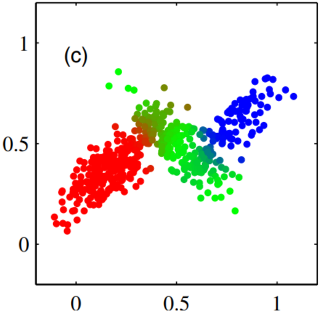
EM: Basic Idea



To compute the responsibilities of a gaussian $\mathcal{N}_j$ for a data point $\boldsymbol{x}_i$ we evaluate the likelihood of $\boldsymbol{x}_i$ being generated by $\mathcal{N}_j$ and normalize it w.r.t. the other components[1]

$$r_{ij} = \frac{\pi_j \mathcal{N}(\boldsymbol{x}_i | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}{\sum_k \pi_k \mathcal{N}(\boldsymbol{x}_i | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}$$





[1] A more theoretical explanation can be found in C. M. Bishop and N. M. Nasrabadi. *Pattern recognition and machine learning*. Vol. 4. 4. Springer, 2006

By using the new responsibilities, we can compute the derivative of the NLL and find the roots to obtain the minimum for each of the parameters:

$$\pi_j^{\text{new}} = \frac{1}{N}\sum_{i=1}^{N} r_{ij} = \frac{N_j}{N}$$

$$\boldsymbol{\mu}_j^{\text{new}} = \frac{1}{N_j}\sum_{i=1}^{N} r_{ij}\boldsymbol{x}_i$$

$$\boldsymbol{\Sigma}_j^{\text{new}} = \frac{1}{N_j}\sum_{i=1}^{N} r_{ij}\left(\boldsymbol{x}_i - \boldsymbol{\mu}_j^{\text{new}}\right)\left(\boldsymbol{x}_i - \boldsymbol{\mu}_j^{\text{new}}\right)^T$$

where

$$N_j = \sum_{i=1}^{N} r_{ij}$$

Online EM

- In Path guiding, the GMMs get refined in an online fashion with a stream of incoming particles.
- Need a formulation of EM that allows for batched updates of the gaussian mixture without storing all previous samples and recomputing the mixture on each update step

General Idea:

- Define a **sufficient statistics** for each mixture.
- Update sufficient statistics with data from the new data point.
- Interpolate new and old data.
- Update model parameters $\boldsymbol{\theta}$ only using the sufficient statistics.

Algorithm 1: Off-line stepwise EM

```
// Index of sufficient statistics
i := 0
repeat
    // Iterate over a batch of N samples
    for q := 0 to N − 1 do
        // E-step: Eq. (4)
        u_i^j := UPDATESUFFICIENTSTATS(s_q, θ^old)
        if i + 1 mod m = 0 then
            // M-step: every m-th observed sample; Eq. (5)
            θ^new := θ̄(u_i^1, ..., u_i^K)
        end
        i := i + 1
    end
until CONVERGED();
```

Figure: Offline EM[a]

[a] Note that for Online EM, the outer loop is removed

The sufficient statistic for a single point is given by a scalar, a vector and a matrix

$$\boldsymbol{u}(\boldsymbol{x}_i) = \left(1, \boldsymbol{x}_i, \boldsymbol{x}_i\boldsymbol{x}_i^T\right)$$

For the j-th Gaussian, the sufficient statistic based on N points is now obtained by averaging

$$\begin{aligned}\boldsymbol{u}_N^j &= \frac{1}{N}\sum_{i=1}^{N} r_{ij}\boldsymbol{u}(\boldsymbol{x}_i)\\ &= \left(\frac{1}{N}\sum_{i=1}^{N} r_{ij}, \frac{1}{N}\sum_{i=1}^{N} r_{ij}\boldsymbol{x}_i, \frac{1}{N}\sum_{i=1}^{N} r_{ij}\boldsymbol{x}_i\boldsymbol{x}_i^T\right)\\ &=: \left((u_r)_N^j, (\boldsymbol{x})_N^j, (\boldsymbol{x}\boldsymbol{x}^T)_N^j\right)\end{aligned}$$

Now, we can rewrite the M-Step equations in terms of the sufficient statistics

$$\pi_j^{\text{new}} = \frac{1}{N}\sum_{i=1}^{N} r_{ij} = (u_r)_N^j$$

$$\boldsymbol{\mu}_j^{\text{new}} = \frac{1}{N_j}\sum_{i=1}^{N} r_{ij}\boldsymbol{x}_i = \frac{(\boldsymbol{x})_N^j}{(u_r)_N^j}$$

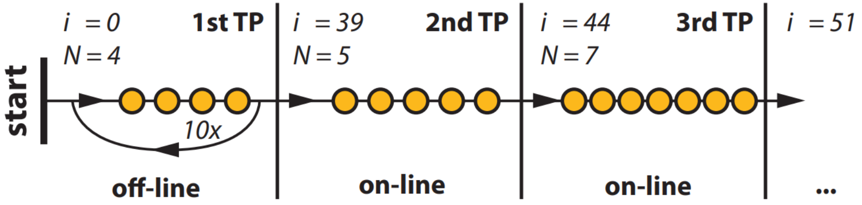
$$\begin{aligned}
\boldsymbol{\Sigma}_j^{\text{new}} &= \frac{1}{N_j}\sum_{i=1}^{N} r_{ij}\left(\boldsymbol{x}_i - \boldsymbol{\mu}_j^{\text{new}}\right)\left(\boldsymbol{x}_i - \boldsymbol{\mu}_j^{\text{new}}\right)^T \\
&= \frac{1}{N_j}\sum_{i=1}^{N} r_{ij}\left[\boldsymbol{x}_i\boldsymbol{x}_i^T - \boldsymbol{x}_i\boldsymbol{\mu}_j^T - \boldsymbol{\mu}_j\boldsymbol{x}_i^T + \boldsymbol{\mu}_j\boldsymbol{\mu}_j^T\right] \\
&= \frac{(\boldsymbol{x}\boldsymbol{x}^T)_N^j - (\boldsymbol{x})_N^j\boldsymbol{\mu}_j^T - \boldsymbol{\mu}_j(\boldsymbol{x}^T)_N^j + (u_r)_N^j\boldsymbol{\mu}_j\boldsymbol{\mu}_j^T}{(u_r)_N^j}
\end{aligned}$$

Note that the update now only consists of the sufficient statistics. The samples $\boldsymbol{x}_i$ are not needed.

Until now, the formulation is equivalent to the standard EM approach. To modify the algorithm to work for a stream of samples $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots\}$, $\boldsymbol{u}_N^j$ is constructed only the first k samples $\boldsymbol{u}_k^j$. New samples are incorporated via linear interpolation:

$$\boldsymbol{u}_k^j = (1 - \eta_k)\boldsymbol{u}_{k-1}^j + \eta_k r_{ij} \boldsymbol{u}(\boldsymbol{x}_i)$$

where $\eta_k = k^{-\alpha}$ is a decreasing sequence with step size $\alpha \in [0.6 : 0.9]$.



- At the beginning, the offline version is iterated until convergence.
- In all successive training passes (TP) each incoming particle is processed once.

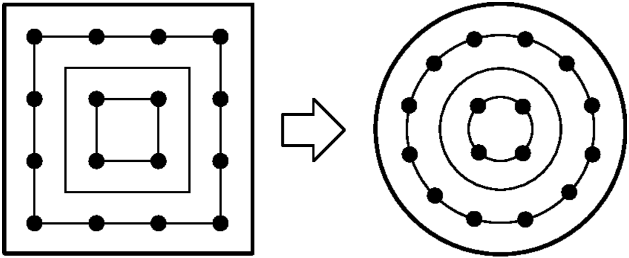
Now, we want to use the presented algorithm in practice:

Questions:

- How to represent the directional distribution $p(\boldsymbol{\Omega}_i|\boldsymbol{x}_i)$
- How to train distributions.
- How to cache distribution and "interpolate" during rendering.

Representing the Distribution

- Mixture of bi-variate Gaussians on a 2D plane
- Project hemisphere $\mathcal{H}_i$ onto unit square using area preserving mapping $\mathcal{S}$ (Shirley and Chiu, 1997)



Trace two kind of distributions:

- Photons traced from the light source $\Rightarrow$ used to guide camera paths.
- Importons traced from the camera $\Rightarrow$ used to guide light paths.
- Enables the use of bi-directional methods

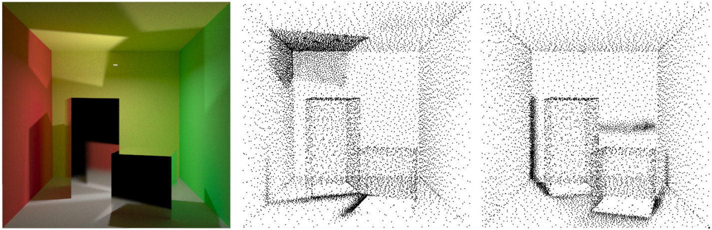


Figure: The black dots represent positions of distributions used for guiding camera (center) and light paths (right).

Spacing of distributions is based on the angular frequency of the radiance/importance function.

Reusing a distribution that is defined at position $\boldsymbol{x}$ at position $\boldsymbol{x}'$ should only be done if the radiance/importance functions are *similar enough*. This will be determined by the **validity radius**

$$r = \frac{1}{\sum_j^K \frac{\pi_j}{r_j}}$$

which describes a weighted harmonic mean of the validity radii of the individual components

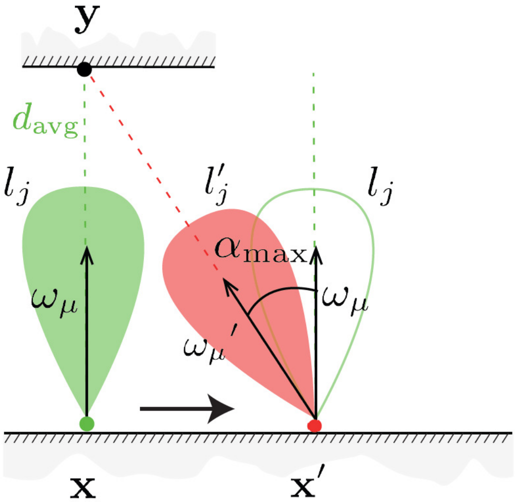
$$r_j \propto \tan(\alpha_{\max})$$

where $\alpha_{\max} = \arccos\big(\big\langle \Omega_\mu, \Omega'_\mu \big\rangle\big)$ is the maximum angle between the mean lobe direction of the original lobe and the lobe at $\boldsymbol{x}'$ that would produce the alleged highlight at $\boldsymbol{y}$.

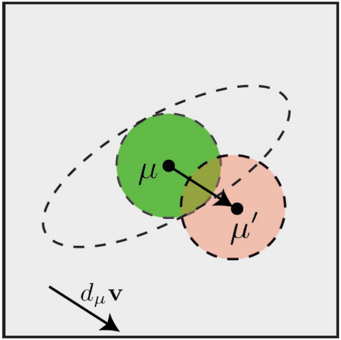
The thresholding is done based on the Kullback-Leibler divergence of the original and the rotated lobe[2].

$$D(P\|Q) = \int_\Omega p(x) \cdot \log \frac{p(x)}{q(x)} \mathrm{d}x$$

[2] J. Vorba, O. Karlık, et al. "On-line Learning of Parametric Mixture Models for Light Transport Simulation–Supplemental material". In: *ACM Transactions on Graphics (TOG)* (2014).

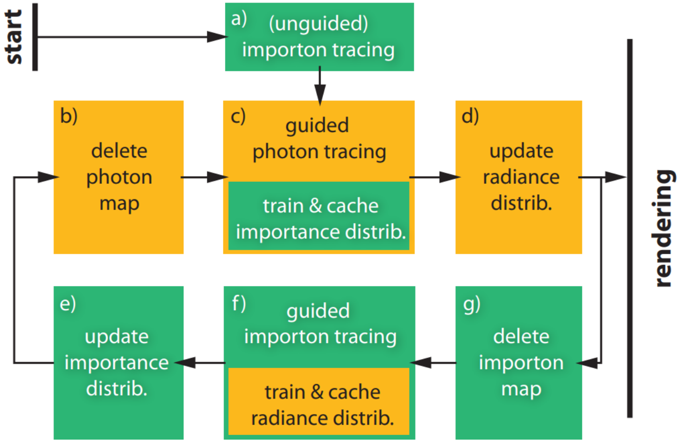


$\mathbf{y}$
d_{avg}
l_j
l'_j
l_j
$\alpha_{\max}$
ω_μ
ω_μ
$\omega_\mu{}'$
$\mathbf{x}$
$\mathbf{x}'$



μ
μ'
$d_\mu \mathbf{v}$

- Before rendering, a training phase is used to fit the distributions.
- A training phase consists of 30 training passes as depicted below.



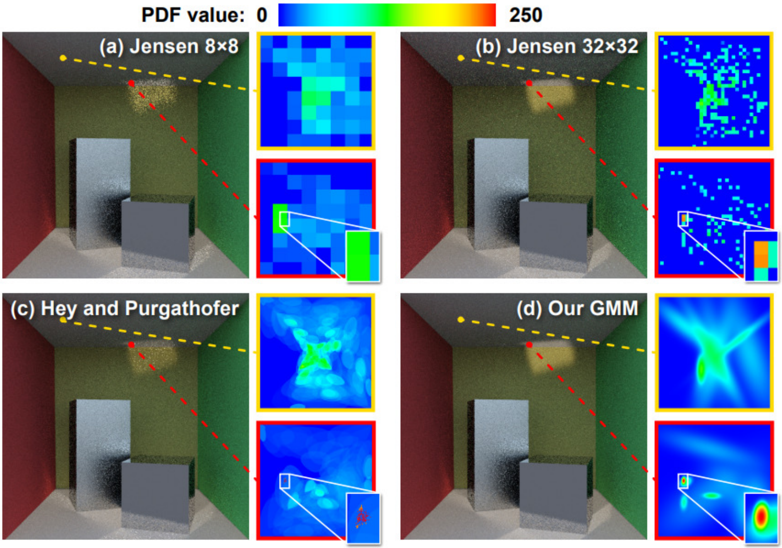
Rendering Pass

For rendering, we need a way to select a fitting distribution

- Query the M nearest distributions at position $\boldsymbol{x}$.
- Check if $\boldsymbol{x}$ lies in the validity radius of the queried distributions.
- Choose the one that minimizes

$$\frac{\|\boldsymbol{x}-\boldsymbol{x}_i\|^2}{h}+2\sqrt{1-\langle\boldsymbol{n},\boldsymbol{n}_i\rangle}$$

  where $\boldsymbol{n}_i$ and $\boldsymbol{x}_i$ are the normal and position of the queried distribution, respectively.
- Randomly sample between BSDF and guided radiance/importance sampling.
- Combine samples using multiple importance sampling.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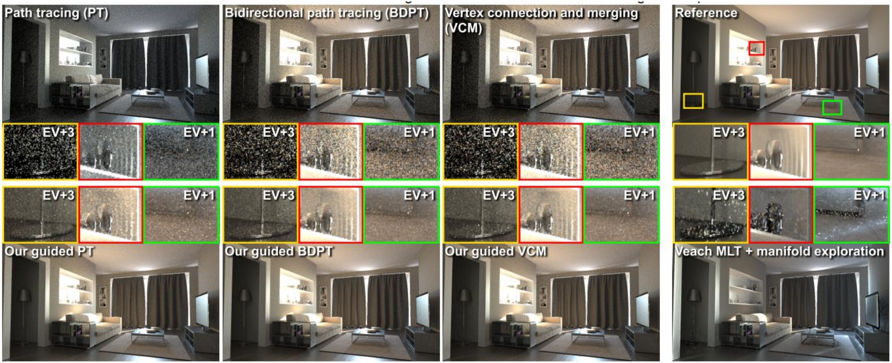
250
(a) Jensen 8×8
(b) Jensen 32×32
(c) Hey and Purgathofer
(d) Our GMM

Results



Path tracing (PT)
Bidirectional path tracing (BDPT)
Vertex connection and merging (VCM)
Reference
EV+3
EV+1
EV+3
EV+1
EV+3
EV+1
EV+3
EV+1
EV+3
EV+1
EV+3
EV+1
EV+3
EV+1
EV+3
EV+1
Our guided PT
Our guided BDPT
Our guided VCM
Veach MLT + manifold exploration

Further Remarks

- Maximum Likelihood tends to overfit. MAP estimates are possible and where also used by the authors.
- When tracing particles, each particle has a weight associated with it. Incorporating this weight into the training leads to an updated M-step of the online algorithm.

Limitations

- For simple scenes, the overhead of querying the cache may offset advantages.
- Fixed number of Gaussian components.
- Only sampling incoming radiance or importance. No good importance sampling technique for glossy-glossy inter-reflections.

Path Guiding for Caustics

- A lot of light phenomena caused by **reflection/refraction**
- Constrained by physical laws $\rightarrow$ difficult to sample
- Making them absent in most renderings.

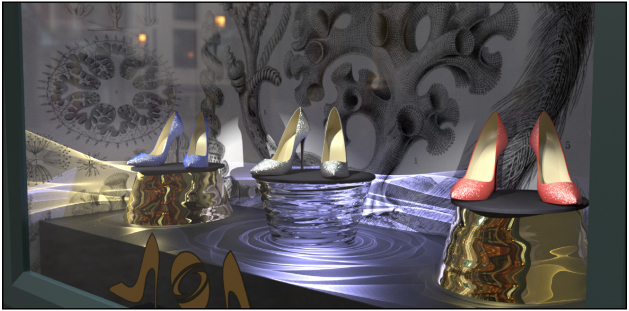
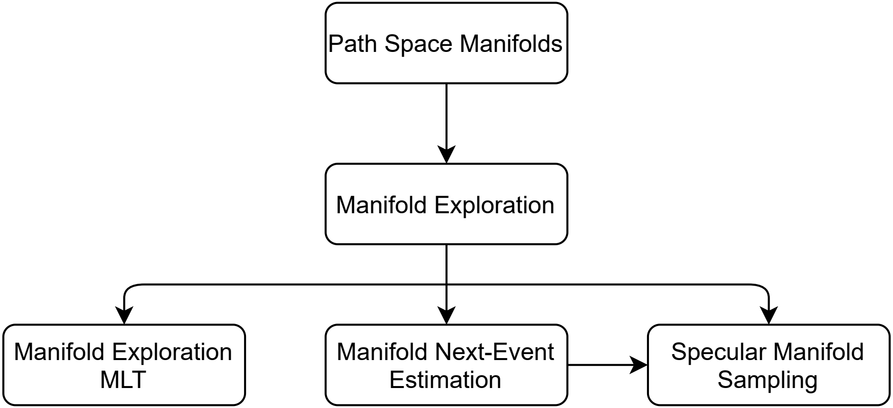


Figure: Shopping Window scene taken from Zeltner et al. (2020)


Path Space Manifolds
Manifold Exploration
Manifold Exploration MLT
Manifold Next-Event Estimation
Specular Manifold Sampling

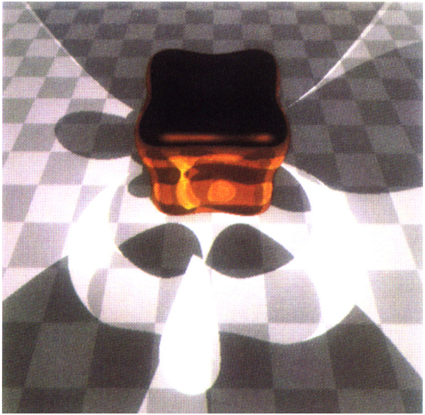


Figure: Curved Reflectors taken from Mitchell and Hanrahan (1992)

- Compute reflected illumination from curved mirrors
- Find extremal paths according to Fermat's principle
- Compute irradiance from Gaussian curvature of geometrical wavefront
- Interval arithmetic

Disadvantages:

- Large intervals
- Too large for systematic pruning of solution space

Figure: Photon Maps taken from Jensen (1996)

- Send photons from light source into scene
- Stored on surface geometry
- Determine illumination by second pass
- Can be incorporated into bidirectional rendering algorithms (Georgiev et al., 2012; Hachisuka et al., 2012)

Disadvantages:

- Memory requirements
- Introduce **blur**

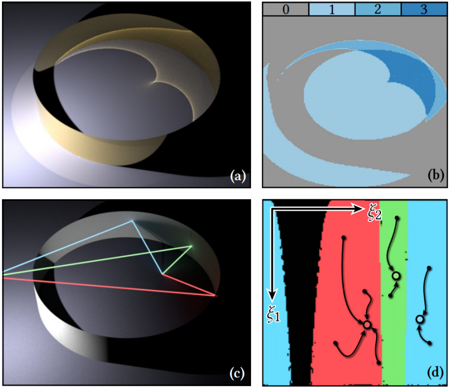


Figure: Manifold Exploration taken from Zeltner et al. (2020)

Manifold Exploration (Jakob and Marschner, 2012)

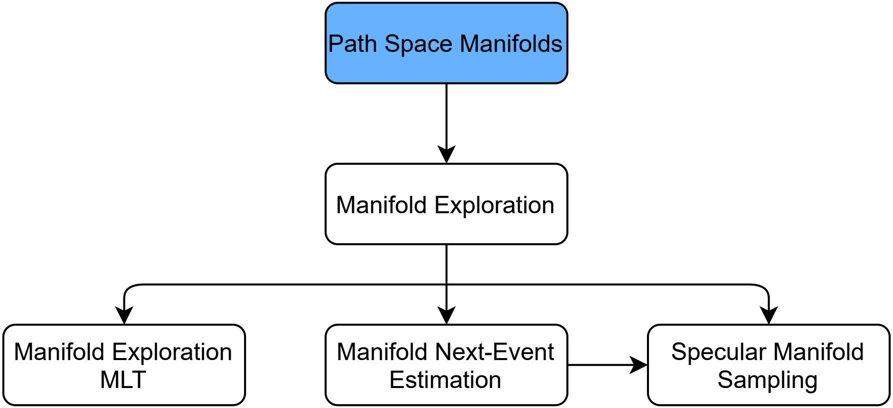
- Use path space manifolds to search specular paths
- Incorporate into Metropolis Light Transport
- New **Perturbation** rule for MLT
- Extendable for volumes

Manifold Next-Event Estimation (Hanika et al., 2015)

- Use Manifold Exploration to find light path connections through refraction
- **Straight-line** initialization

Specular manifold sampling (Zeltner et al., 2020)

- Extension of MNEE
- **Random** initialization
- Unbiased and biased algorithm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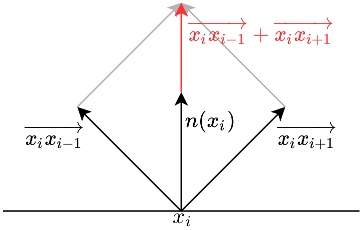
Path Space: Set of all paths in a scene

$$\mathcal{P} = \bigcup_{n=2}^{\infty} \mathcal{P}_n$$

$$\mathcal{P}_n = \{\boldsymbol{x}_1 \ldots \boldsymbol{x}_n \,|\, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathcal{M}\}$$

Specular manifold: Subset of paths fulfilling specular constraints:

$$(\overrightarrow{\boldsymbol{x}_i \boldsymbol{x}_{i-1}} + \overrightarrow{\boldsymbol{x}_i \boldsymbol{x}_{i+1}}) \parallel \boldsymbol{n}(\boldsymbol{x}_i)$$



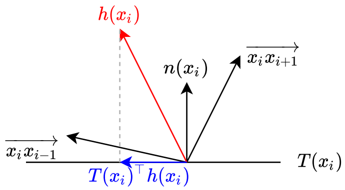
Jakob and Marschner, 2012

Formulate constraint by projecting the half vector $\boldsymbol{h}$ into the local tangent plane $\boldsymbol{T}(\boldsymbol{x}_i)$:

$$\boldsymbol{c}_i(\boldsymbol{x}_{i-1}, \boldsymbol{x}_i, \boldsymbol{x}_{i+1}) = \boldsymbol{T}(\boldsymbol{x}_i)^\top \boldsymbol{h}(\boldsymbol{x}_i, \overrightarrow{\boldsymbol{x}_i\boldsymbol{x}_{i-1}}, \overrightarrow{\boldsymbol{x}_i\boldsymbol{x}_{i+1}}) = \boldsymbol{0}$$

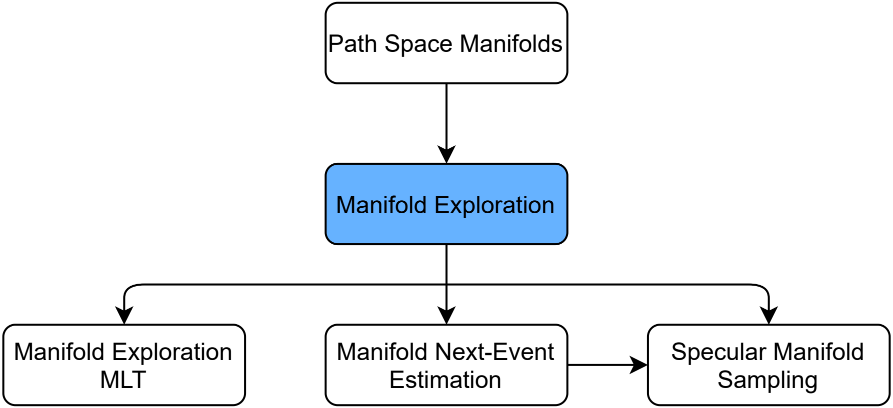
Stack constraints of each specular vertex $\boldsymbol{C} : \mathbb{R}^{2n} \to \mathbb{R}^{2p} \Rightarrow$ Implicit definition of the specular manifold:

$$\mathcal{S} = \{\bar{\boldsymbol{x}} \mid \boldsymbol{C}(\bar{\boldsymbol{x}}) = \boldsymbol{0}\}$$



Jakob and Marschner, 2012

Outline



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Theorem (Implicit Function Theorem)

Let $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ *be a continuously differentiable function and let* $\mathbb{R}^n \times \mathbb{R}^m$ *have coordinates* $(\boldsymbol{x}, \boldsymbol{y}) = (x_1, \ldots, x_n, y_1, \ldots, y_m)$*. Fix a point* $(\boldsymbol{a}, \boldsymbol{b})$ *with* $f(\boldsymbol{a}, \boldsymbol{b}) = 0$*. If the Jacobian matrix*

$$\boldsymbol{J}_{f,\boldsymbol{y}}(\boldsymbol{a}, \boldsymbol{b}) = \left[\frac{\partial f_i}{\partial y_j}(\boldsymbol{a}, \boldsymbol{b})\right]$$

is invertible, then there exists an open set $U \subset \mathbb{R}^n$ *containing* $\boldsymbol{a}$ *such that there exists a unique continuously differentiable function* $g : U \to \mathbb{R}^m$ *such that* $g(\boldsymbol{a}) = \boldsymbol{b}$ *and* $f(\boldsymbol{x}, g(\boldsymbol{x})) = 0$ *for all* $\boldsymbol{x} \in U$*. Moreover, denoting the left-hand panel of the Jacobian matrix*

$$\boldsymbol{J}_{f,\boldsymbol{x}}(\boldsymbol{a}, \boldsymbol{b}) = \left[\frac{\partial f_i}{\partial x_j}(\boldsymbol{a}, \boldsymbol{b})\right],$$

the Jacobian matrix of the partial derivatives of g *in* U *is given by the matrix product*

$$\left[\frac{\partial g_i}{\partial x_j}(\boldsymbol{x})\right]_{m \times n} = -\left[\boldsymbol{J}_{f,\boldsymbol{y}}(\boldsymbol{x}, g(\boldsymbol{x}))\right]^{-1}_{m \times m} \left[\boldsymbol{J}_{f,\boldsymbol{x}}(\boldsymbol{x}, g(\boldsymbol{x}))\right]_{m \times n}$$

Example:
Consider a 2D circle given by

$$f(x, y) = x^2 + y^2 - 1$$

Consider the Jacobian

$$\boldsymbol{J}_f(x, y) = (2x, 2y)$$

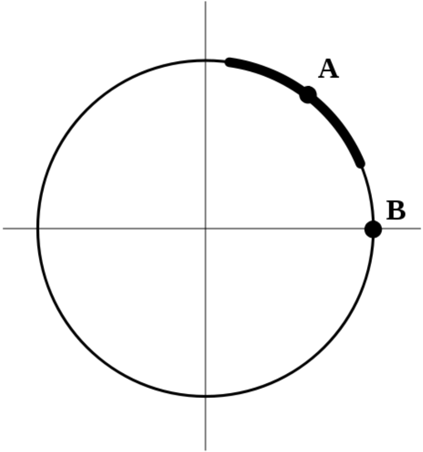
Although we do not know the explicit function $g(x)$, we can calculate the derivative w.r.t. x as

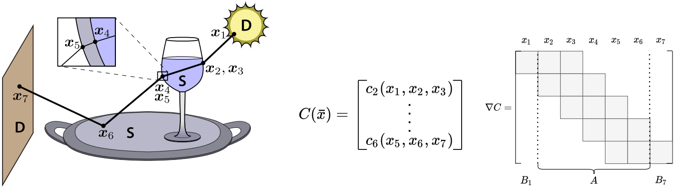
$$\frac{\mathrm{d}g(x)}{\mathrm{d}x} = -\frac{\partial f(x, y)}{\partial y} \cdot 2x = -\frac{x}{y}.$$

We can verify this by using the explicit functions

$$g(x) = \pm\sqrt{1 - x^2}$$

Note that for $y = 0$ there does not exist such an explicit function because in this case, $\boldsymbol{J}_{f,\boldsymbol{y}}$ is non-invertible.





(a) An example path

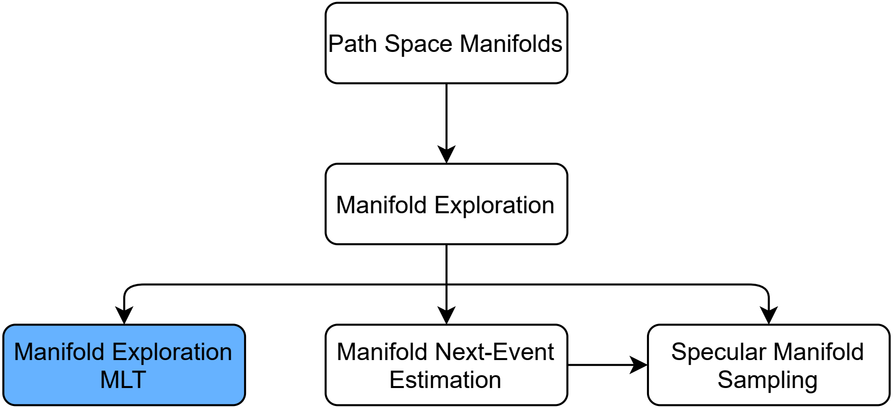
(b) The constraint function

(c) The corresponding Jacobian

Find local parameterization of the specular manifold using the **Implicit Function Theorem** w.r.t. the start and end point:

$$\boldsymbol{T}_{\mathcal{S}}(\bar{\boldsymbol{x}}) = -\boldsymbol{A}^{-1}[\boldsymbol{B}_1\ \boldsymbol{B}_n]$$

Jakob and Marschner, 2012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Manifold Exploration
Manifold Exploration MLT
Manifold Next-Event Estimation
Specular Manifold Sampling

Question: How to incorporate Manifold Exploration into rendering algorithms?

Metropolis Light Transport[3]**:**

- **Idea:** Define Markov Chain that has a stationary distribution proportional to the function to be integrated (i.e. light contribution of each ray $f_j(\bar{\boldsymbol{x}})$)
- Define a transition rule $\bar{\boldsymbol{x}} \rightarrow \bar{\boldsymbol{x}}'$ based on a proposal distribution $T(\bar{\boldsymbol{x}}, \bar{\boldsymbol{x}}')$
- Accept new state with acceptance probability

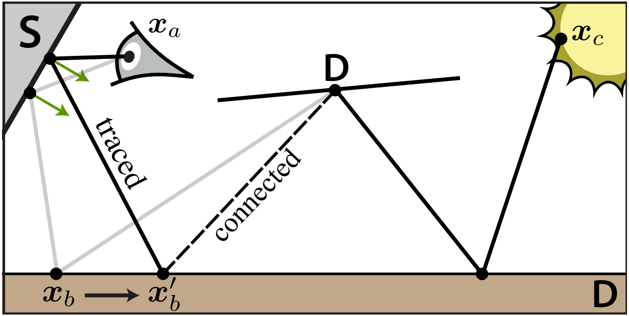
$$r(\bar{\boldsymbol{x}}, \bar{\boldsymbol{x}}') = \min\left(1, \frac{f_j(\bar{\boldsymbol{x}}')T(\bar{\boldsymbol{x}}', \bar{\boldsymbol{x}})}{f_j(\bar{\boldsymbol{x}})T(\bar{\boldsymbol{x}}, \bar{\boldsymbol{x}}')}\right)$$

Question: What is a good transition rule?
In principle every transition rule is valid but rules with low acceptance probability waste computation time.

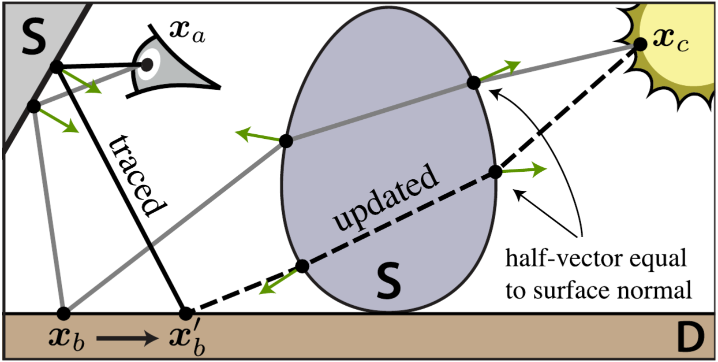
Idea: Use a **perturbation** rule that modifies the current path only slightly.

[3] E. Veach and L. J. Guibas. "Metropolis light transport". In: *Proceedings of the 24th annual conference on Computer graphics and interactive techniques*. 1997, pp. 65–76.

- **Sampling** and **Connection** step (Veach and Guibas, 1997).
- Given a path $\bar{x}$ sample three diffuse vertices $\boldsymbol{x}_a, \boldsymbol{x}_b, \boldsymbol{x}_c$.
- Sample a new outgoing direction and follow it starting from $\boldsymbol{x}_a$ until a point $\boldsymbol{x}_b'$ in the neighborhood of $\boldsymbol{x}_b$ is found.
- Connect to old path through "DD" connection edge.

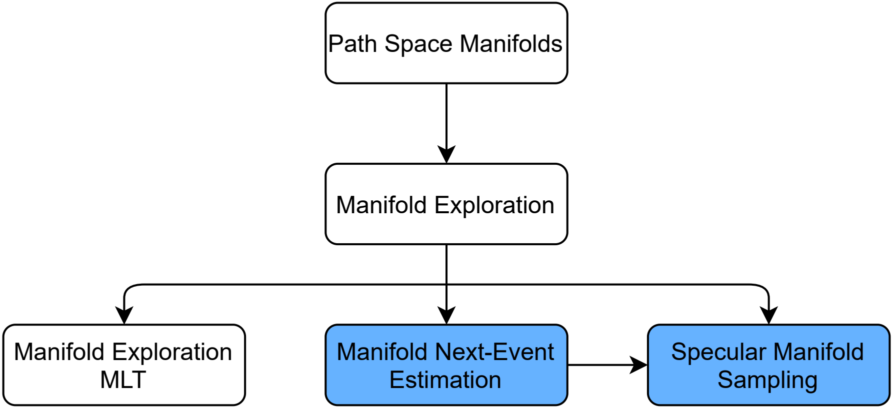


Now: Incorporate Manifold Exploration[4] into perturbation strategy.
Connect x'_b and x'_c via the `ManifoldWalk` algorithm.



[4] W. Jakob and S. Marschner. "Manifold exploration: a Markov Chain Monte Carlo technique for rendering scenes with difficult specular transport". In: *ACM Transactions on Graphics (TOG)* 31.4 (2012), pp. 1–13.

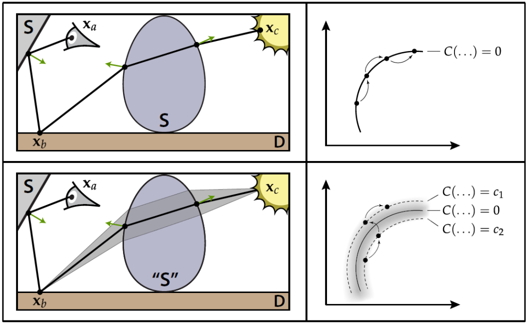
Outline



Path Space Manifolds
Manifold Exploration
Manifold Exploration MLT
Manifold Next-Event Estimation
Specular Manifold Sampling

Specular

Glossy



Offset Manifold walk:

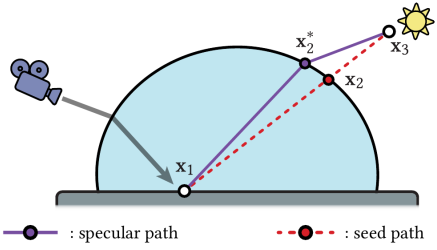
- Define offset manifold $\mathcal{S}_{\boldsymbol{o}} = \{\bar{\boldsymbol{x}} \,|\, \boldsymbol{C}(\bar{\boldsymbol{x}}) = \boldsymbol{o}\}$
- Sample a microfacet normal from normal map
- Keep sampled normal invariant when doing the manifold walk.

Jakob, 2013

Next-Event Estimation: Split indirect and direct illumination by establishing a connection to the light source.
Problem: Light and surface can be separated by refractive surface $\Rightarrow$ Straight line is not a valid path.
Solution: Use `ManifoldWalk` algorithm to transform straight line initialization into valid path.
Needed for Monte Carlo estimator $\frac{f(x)}{p(x)}$: Probability of sampling the path $\Rightarrow$ Marginalization over path space:

$$p(\bar{\boldsymbol{x}}^+) = \int_{\mathcal{P}} p(\bar{\boldsymbol{x}}^+ \mid \bar{\boldsymbol{x}}_0) p(\bar{\boldsymbol{x}}_0) \mathrm{d}\bar{\boldsymbol{x}}_0$$

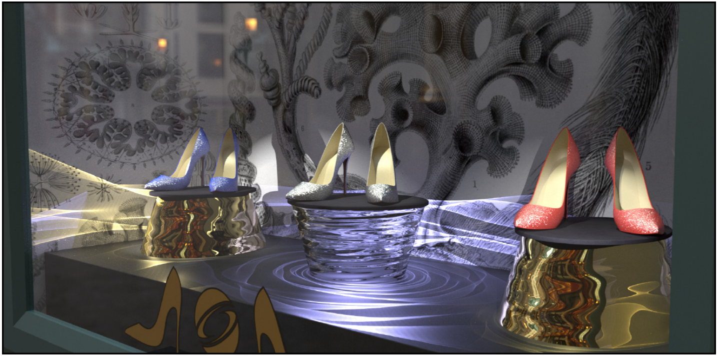
Evaluation of integral becomes trivial as $p(\bar{\boldsymbol{x}}_0)$ is a delta distribution.



Hanika et al., 2015

Hanika et al., 2015

- Many light effects produced by superposition of different specular light paths ⇒ Cannot be reconstructed by MNEE



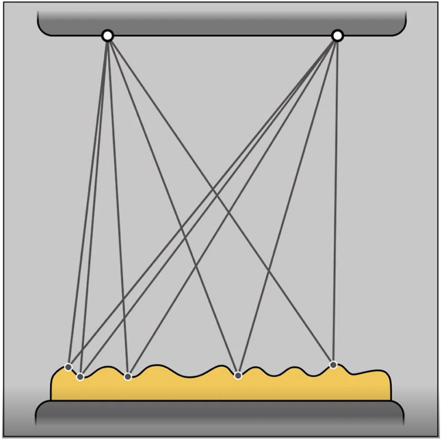
Zeltner et al., 2020

Basic approach of Specular Manifold Sampling[5]

Sample random solutions

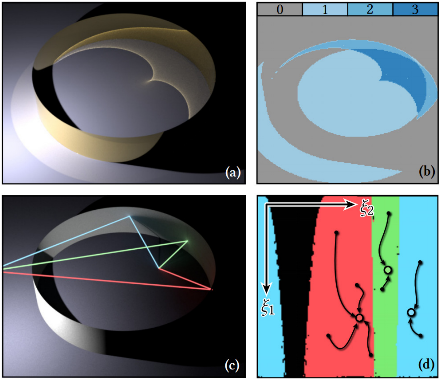
- Sample seed points, e.g. uniformly
- Connect with endpoints
- Run Newton solver to get $\bar{\boldsymbol{x}}$
- Compute radiance

$$L = \int f(\bar{\boldsymbol{x}}) d\bar{\boldsymbol{x}}$$
$$= \frac{1}{N} \sum_{i=1}^{n} \frac{f(\bar{\boldsymbol{x}}_i)}{p(\bar{\boldsymbol{x}}_i)}$$



[5] T. Zeltner et al. "Specular manifold sampling for rendering high-frequency caustics and glints". In: *ACM Transactions on Graphics (TOG)* 39.4 (2020), pp. 149–1.

- Use random initialization
- How to compute probability of finding the chosen path $\Rightarrow$ Specular manifold sampling
- Independent on sampling strategy: sample $\boldsymbol{\xi} = (\xi_1, \xi_2) \in [0,1)^2 =: \mathcal{U}$ and transform them into the given sampling strategy.
- Sample space $\mathcal{U}$ divided into **basins of convergence** $\mathcal{B}_k$

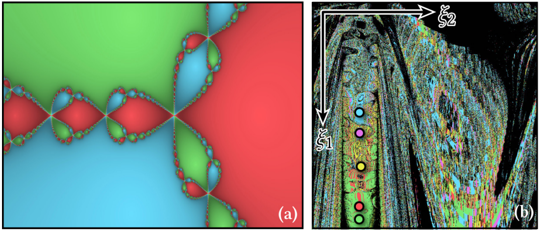


Zeltner et al., 2020

- $p(\boldsymbol{x}_k)$ given by area of basins $\Rightarrow p(\boldsymbol{x}_k) = \int_{\mathcal{U}} 1_{\mathcal{B}_k}(\boldsymbol{\xi})\mathrm{d}\boldsymbol{\xi}$
- Calculation of area difficult because of complex structures in the sample space.

Simple solution: Sample i.i.d. $\boldsymbol{\xi} \in \mathcal{U}$ and estimate the area?

$$\langle p_k \rangle = \frac{1}{N} \sum_{i=1}^{N} 1_{\mathcal{B}}(\boldsymbol{\xi})$$



Zeltner et al., 2020

This method introduces non-zero bias:

- $p(\boldsymbol{x}_k)$ appears in the **denominator** of the Monte Carlo estimator $\frac{f(\bar{\boldsymbol{x}})}{p(\bar{\boldsymbol{x}})}$ but $\mathbb{E}[1/x] \neq 1/\mathbb{E}[x]$
- Move estimation from denominator to numerator by using a **geometric series** (Booth, 2007):

$$\frac{1}{p_k} = \frac{1}{\int_{\mathcal{U}} 1_{\mathcal{B}_k}(\boldsymbol{\xi})\mathrm{d}\boldsymbol{\xi}} = \frac{1}{1-a} = \sum_{i=1}^{\infty} a^i$$

with $a = 1 - \int_{\mathcal{U}} 1_{\mathcal{B}_k}(\boldsymbol{\xi})\mathrm{d}\boldsymbol{\xi}$

Unbiased specular manifold sampling:

- Run `ManifoldWalk` once.
- Repeat walks until the same solution is found again $\Rightarrow$ number of attempts $n_k = \langle 1/p_k \rangle$

Zeltner et al., 2020

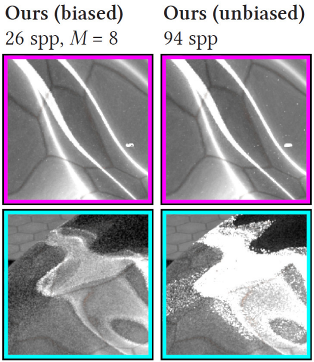
- May find many solutions in second phase of `ManifoldWalk`.
- Are not further considered.
- How to integrate this information into the estimator?
- Difficult to do without bias

This leads to the **biased specular manifold sampling** algorithm:

- Introduce fixed budget M
- Cluster M samples into a set of unique solutions.
- Estimate Path throughput at shading point $\boldsymbol{x}_2$:

$$\frac{1}{M}\sum_{l=1}^{L} n_l \frac{f(\boldsymbol{x}_2^{(l)})}{p(\boldsymbol{x}_2^{(l)})} \approx \frac{1}{M}\sum_{l=1}^{L} n_l f(\boldsymbol{x}_2^{(l)})\frac{M}{n_l} = \sum_{l=1}^{L} f(\boldsymbol{x}_2^{(l)})$$

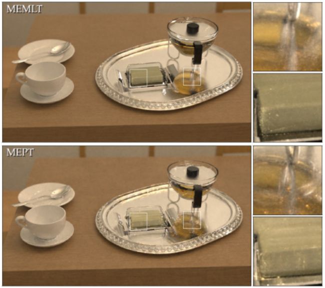
- Reduces variance.
- Leads to energy loss.



Zeltner et al., 2020

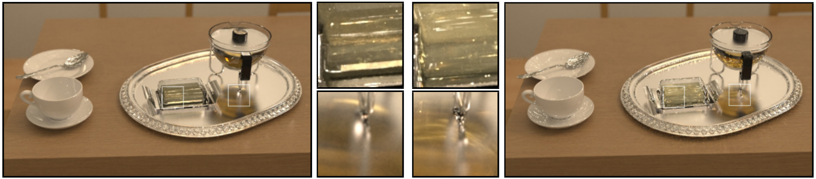
Manifold Exploration Path Tracing (MEPT) implemented in Mitsuba (Jakob and Marschner, 2012).
Compared against:

- Metropolis Light Transport (MLT) (Veach and Guibas, 1997).
- Primary Sample Space MLT (PSSMLT) (Kelemen et al., 2002).
- Energy redistribution path tracing (ERPT) (Cline et al., 2005).

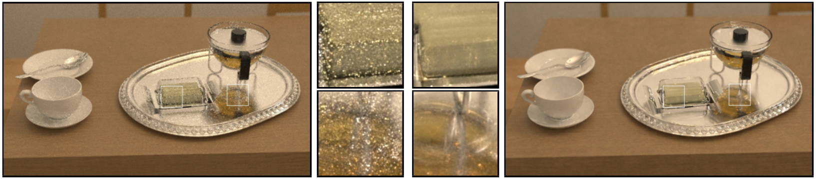


Specular manifold sampling implemented in Mitsuba 2 (Zeltner et al., 2020).
Compared against:

- Conventional Path tracing.
- Manifold Next-Event Estimation (Hanika et al., 2015).
- Comparisons between unbiased and biased variant.



(a) MLT (b) ERPT



(c) PSSMLT (d) MEPT

Figure: Equal time renderings (1 hour) comparing the different approaches taken from Jakob and Marschner (2012).

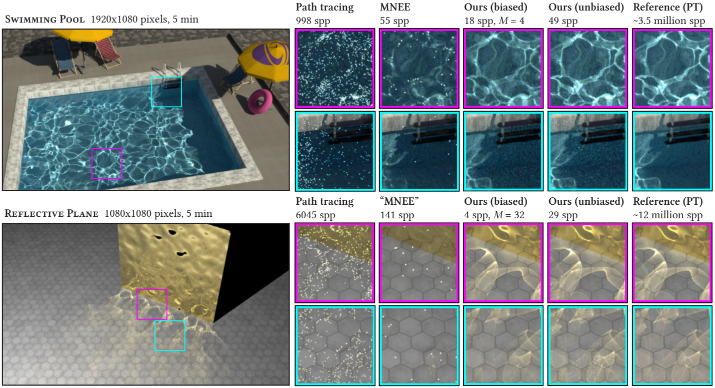


Figure: Equal time comparisons (5 minutes) comparing MNEE to the unbiased and biased SMS algorithm taken from Zeltner et al. (2020).

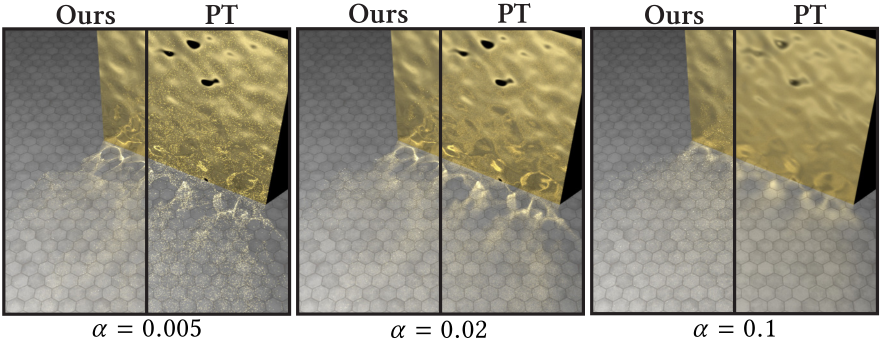


Figure: Rough reflective plane with rough metallic surface with Beckmann normal distribution taken from Zeltner et al. (2020).

Conclusion

- A lot of light effects emerge from specular interactions.
- Difficult to sample as they are constrained by physical laws.
- Systematically explore path space via the specular manifold.
- Incorporate Manifold Exploration into MLT
- MNEE: Lightweight method for finding specular paths outside MCMC context.
- SMS: Use random initialization to find specular paths
- Many extensions: Offset manifolds, volumes, Normal mapped surfaces,. . .
- How to extend to different path configurations?
- How to incorporate into MIS strategies?

[1] C. M. Bishop and N. M. Nasrabadi. *Pattern recognition and machine learning*. Vol. 4. 4. Springer, 2006.

[2] T. E. Booth. "Unbiased Monte Carlo estimation of the reciprocal of an integral". In: *Nuclear science and engineering* 156.3 (2007), pp. 403–407.

[3] D. Cline, J. Talbot, and P. Egbert. "Energy redistribution path tracing". In: *ACM Transactions on Graphics (TOG)* 24.3 (2005), pp. 1186–1195.

[4] I. Georgiev, J. Krivanek, T. Davidovic, and P. Slusallek. "Light transport simulation with vertex connection and merging.". In: *ACM Trans. Graph.* 31.6 (2012), pp. 192–1.

[5] T. Hachisuka, J. Pantaleoni, and H. W. Jensen. "A path space extension for robust light transport simulation". In: *ACM Transactions on Graphics (TOG)* 31.6 (2012), pp. 1–10.

[6] J. Hanika, M. Droske, and L. Fascione. "Manifold next event estimation". In: *Computer graphics forum*. Vol. 34. 4. Wiley Online Library. 2015, pp. 87–97.

[7] W. Jakob. *Light transport on path-space manifolds*. Cornell University Dissertation, 2013.

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[11] D. Mitchell and P. Hanrahan. "Illumination from curved reflectors". In: *Proceedings of the 19th annual conference on Computer graphics and interactive techniques*. 1992, pp. 283–291.

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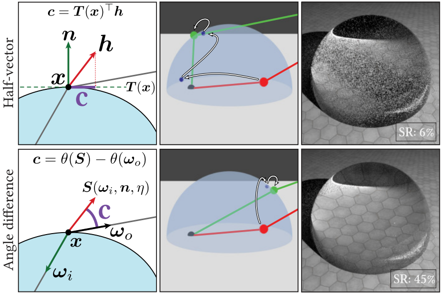
[14] J. Vorba, O. Karlik, M. Sik, T. Ritschel, and J. Krivanek. "On-line learning of parametric mixture models for light transport simulation". In: *ACM Transactions on Graphics (TOG)* 33.4 (2014), pp. 1–11.

[15] J. Vorba, O. Karlık, M. Šik, T. Ritschel, and J. Křivánek. "On-line Learning of Parametric Mixture Models for Light Transport Simulation–Supplemental material". In: *ACM Transactions on Graphics (TOG)* (2014).

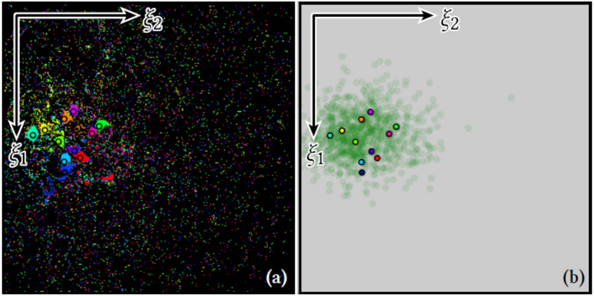
[16] T. Zeltner, I. Georgiev, and W. Jakob. "Specular manifold sampling for rendering high-frequency caustics and glints". In: *ACM Transactions on Graphics (TOG)* 39.4 (2020), pp. 149–1.



- Compared to other methods ⇒ Manifold Exploration has to do large steps.
- Constraints may lead to back facing solutions
- Improve constraints by measuring angle distance

$$c_i = \begin{pmatrix} \theta(S(\overrightarrow{x_i x_{i-1}}, n_i, \eta_i)) - \theta(\overrightarrow{x_i x_{i+1}}) \\ \phi(S(\overrightarrow{x_i x_{i-1}}, n_i, \eta_i)) - \phi(\overrightarrow{x_i x_{i+1}}) \end{pmatrix}$$

Zeltner et al., 2020



Two-stage Manifold walk:

- Do a manifold walk using the smooth geometry
- Sample a microfacet normal from normal map
- Do a second manifold walk using the first solution as seed path.

Zeltner et al., 2020